

TECHNICAL ARCHITECTURE SPEC

NEXT.JS 14+ APP ROUTER

TYPESCRIPT STRICT

TAILWIND CUSTOM PROPS

TECHNICAL REQUIREMENTS DOCUMENT (TRD)

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ARCHITECTURE

Server-First Hybrid (RSC)

DESIGN ENGINE

4 CSS Custom Property Themes

I18N & DIRECTION

next-intl (LTR & RTL)

EDGE TARGET

Vercel Global CDN

1 Component Hierarchy & Server Boundary Map

Next.js 14+ App Router is designed to minimize client bundle payloads. Static presentation components remain pure Server Components (RSC), while client interactivity is strictly isolated into dedicated islands:

COMPONENT LAYER	TYPE	FILE LOCATION	TECHNICAL RESPONSIBILITY
<code>[locale]/layout.tsx</code>	Server Component	<code>src/app/[locale]/layout.tsx</code>	Injects HTML root, locale attributes (<code>lang</code> , <code>dir</code>), anti-flicker theme script, and font classes.
<code>Header.tsx</code>	Client Component	<code>src/components/layout/Header.tsx</code>	Sticky navigation bar, responsive mobile hamburger drawer, active link observer.
<code>ThemeToggle.tsx</code>	Client Component	<code>src/components/ui/ThemeToggle.tsx</code>	Switches <code>data-theme</code> on <code>document.documentElement</code> ; synchronizes with <code>localStorage</code> .
<code>LanguageToggle.tsx</code>	Client Component	<code>src/components/ui/LanguageToggle.tsx</code>	Route-based locale switcher (<code>/en</code> , <code>/ta</code> , <code>/ur</code>) powered by <code>next-intl</code> router.
<code>HeroSection.tsx</code>	Server / Client Hybrid	<code>src/components/sections/HeroSection.tsx</code>	Static title and badge SSR; client island for <code>TypedText.tsx</code> animated cycling headline.
<code>ProjectsSection.tsx</code>	Client Component	<code>src/components/sections/ProjectsSection.tsx</code>	Grid of 4 projects featuring 3D hover perspective tilt and category filter toggles.
<code>ContactForm.tsx</code>	Client Component	<code>src/components/ui/ContactForm.tsx</code>	Controlled inputs, client-side regex validation, async POST to Formspree API, toast feedback.

2 Data Contracts & TypeScript Interfaces

All data entities are governed by strict TypeScript interfaces in `src/types/index.ts` to guarantee type safety and compile-time verification:

```
// src/types/index.ts
export type ThemeMode = 'glassmorphism' | 'minimal' | 'warm' | 'terminal';
export type Locale = 'en' | 'ta' | 'ur';

export interface Project {
  slug: string;
  title: string;
  subtitle: string;
  category: 'Full-Stack Web' | 'Mobile / Cross-Platform' | 'QA & Backend';
  featured: boolean;
  status: 'Live on Vercel' | 'Play Store Bundle' | 'Production Ready' | 'Testing Suite';
  statusColor: 'emerald' | 'indigo' | 'amber' | 'blue';
  description: string;
  problemStatement: string;
  architectureDetails: string[];
  keyFeatures: string[];
  techStack: string[];
  liveUrl?: string;
  githubUrl?: string;
  mockupImage: string;
  interviewDefenseNotes: {
    architecturalChoice: string;
    tradeoffConsidered: string;
    failureHandling: string;
  };
};

export interface SkillItem {
  name: string;
  category: 'Languages' | 'Frontend' | 'Backend & APIs' | 'Mobile' | 'Databases & Cloud' | 'QA & DevOps';
  proficiencyLevel: 'Proficient' | 'Familiar' | 'Academic';
  iconKey: string;
}

export interface ContactFormData {
  fullName: string;
  email: string;
  subject?: string;
  message: string;
}
```

3 The 4-Theme CSS Custom Property Engine

Themes are implemented entirely through CSS custom properties attached to `html[data-theme="..."]`. This guarantees zero DOM reflows and zero component re-renders when switching themes.

CSS VARIABLE	GLASSMORPHISM DARK (DEFAULT)	MINIMAL LIGHT	WARM EARTHY	TERMINAL HACKER
--bg-primary	#030712 (Slate-950)	#ffffff (Pure White)	#faf8f5 (Warm Alabaster)	#050505 (Deep Black)
--bg-card	rgba(255, 255, 255, 0.05)	#ffffff (White)	#ffffff (Off-White)	#0d1117 (GitHub Dark)
--border-card	rgba(255, 255, 255, 0.12)	#e2e8f0 (Slate-200)	#e7dfd5 (Stone-300)	#00ff41 (Matrix Green)
--text-primary	#f8fafc (Slate-50)	#0f172a (Slate-900)	#2d3748 (Charcoal)	#00ff41 (Phosphor Green)
--accent-primary	#6366f1 (Indigo-500)	#2563eb (Blue-600)	#c2684e (Terracotta)	#00ff41 (Matrix Green)
--font-family	Plus Jakarta Sans	Plus Jakarta Sans	Plus Jakarta Sans	JetBrains Mono

Flash of Unstyled Content (FOUC) Guard

To prevent a blinding white flash when a returning visitor opens the site in dark mode, an inline script in `<head>` immediately evaluates `localStorage.getItem('safith_portfolio_theme')` and sets `document.documentElement.setAttribute('data-theme', theme)` before the browser paints the first pixel.

4 Internationalization (i18n) & RTL Architecture

The application integrates `next-intl` with App Router middleware to handle localized routing (`/[locale]/...`) and automatic Right-to-Left (RTL) mirroring for Urdu:

LOCALE	LANGUAGE	DIRECTION (DIR)	FONT FALLBACK STACK	LAYOUT RULES ENFORCED
en	English (Default)	ltr	Plus Jakarta Sans, Inter, sans-serif	Standard left-to-right flex/grid flows.
ta	Tamil	ltr	Noto Sans Tamil, sans-serif	Increased line-height (1.65) for Tamil glyph ascenders/descenders.
ur	Urdu	rtl	Noto Nastaliq Urdu, sans-serif	Reversed margins (<code>ms-*</code> , <code>me-*</code>), mirrored icons, flipped chevrons.

5 Motion Choreography & Performance Budget

All page transitions utilize Framer Motion with CSS hardware acceleration to maintain a steady 60 FPS on mobile and low-power hardware:

Motion Animation Tokens

Fade In Up: `opacity: [0, 1]`, `y: [24, 0]` (Duration: 0.5s, Ease: Cubic-Bezier)

Card Hover Tilt: `scale: 1.015`, `y: -6px`, Box-shadow elevation glow

Typing Cyclcr: Character delay: 60ms, Word pause: 2000ms, Backspace delay: 35ms

Reduced Motion Guard: Honors OS `prefers-reduced-motion: reduce` setting

Performance & Core Web Vitals Budget

Largest Contentful Paint (LCP): < 1.2s (Optimized WebP images via `next/image`)

Cumulative Layout Shift (CLS): 0.00 (Zero layout jumping via reserved aspect ratios)

First Input Delay / INP: < 50ms (Zero heavy blocking client JavaScript)

Total Bundle Size: < 95 KB gzip first-load JS transferred

6 Serverless Integration & Quality Assurance

The contact module integrates with Formspree's headless email API, eliminating the need for a self-hosted database or server process while maintaining enterprise-grade spam protection.

QUALITY VECTOR	AUTOMATED COMMAND	ACCEPTANCE STANDARD
TypeScript Strict Mode	<code>npx tsc --noEmit</code>	Exit code 0, 0 type errors across all routes and components.
ESLint Rules	<code>npm run lint</code>	0 errors, 0 warnings; enforcement of Next.js best practices.
Production Build	<code>npm run build</code>	Static HTML generated for all routes; zero bundle bloat.
Lighthouse Audit	Headless Lighthouse	Performance: ≥ 90, Accessibility: ≥ 95, SEO: ≥ 95, Best Practices: ≥ 95.